

FoveaDrive: Vehicle-Aware Adaptive 2.5D Mapping System

Software Requirements Specification (SRS)

1. Introduction

1.1 Purpose

This document defines the software requirements for FoveaDrive, a real-time adaptive mapping system that converts LiDAR-like data into a variable-resolution 2.5D map and evaluates terrain drivability based on vehicle parameters.

1.2 Scope

FoveaDrive demonstrates a foveated perception system where:

- High-resolution mapping is applied near the vehicle
- Low-resolution mapping is used for distant regions
- Terrain is classified as drivable or non-drivable based on vehicle characteristics

This is a prototype system designed for hackathon demonstration using:

- Simulated LiDAR datasets (primary)
- Ultrasonic + servo hardware (proof-of-concept)

2. Overall Description

2.1 Product Perspective

The system consists of:

- Data input (LiDAR dataset / ultrasonic scan)
- Processing engine (adaptive grid + classification)
- Vehicle calibration module
- Web-based visualization dashboard

2.2 Product Functions

The system will:

1. Load and process point cloud data
2. Convert 3D points into a 2.5D adaptive grid

3. Classify terrain and obstacles
4. Allow vehicle parameter customization
5. Determine drivability per grid cell
6. Display results in real-time on a web dashboard

2.3 User Classes

- Judges (primary users)
- Developers (team members)

3. System Features

3.1 Adaptive Resolution Mapping (Core Feature)

Divide environment into distance-based zones with different grid resolutions:

Distance Range	Resolution
0–10 m	High (5 cm)
10–30 m	Medium
30–100 m	Low (50 cm)

3.2 2.5D Grid Representation

Each grid cell stores:

- Height (average Z value)
- Terrain label (drivable / obstacle)

3.3 Terrain Classification (Simplified)

- Ground: low height variation
- Obstacles: high elevation
- Dynamic objects: simulated or dataset labels

3.4 Vehicle Calibration (USP Feature)

Users can input:

- Ground clearance
- Wheel radius
- Vehicle width

Predefined profiles:

- Sedan

- SUV
- Truck

3.5 Drivability Engine

The system evaluates terrain using:

- Obstacle height vs ground clearance
- Surface smoothness

Output:

- Drivable: Green
- Non-drivable: Red

3.6 Real-Time Visualization Dashboard

The UI will:

- Display adaptive grid map
- Show color-coded terrain
- Update based on vehicle selection
- Provide parameter sliders

3.7 Hardware Simulation Module

- Ultrasonic sensor mounted on servo
- Rotational scanning (0–180°)
- Convert polar data to Cartesian points
- Used as proof-of-concept for LiDAR

4. External Interface Requirements

4.1 User Interface (Web Dashboard)

Components:

- Map display (canvas/grid view)
- Vehicle selection dropdown
- Parameter sliders
- Status panel (FPS, mode)

4.2 Hardware Interface

- Ultrasonic sensor
- Servo motor
- Microcontroller (Arduino/ESP32)

4.3 Software Interface

- Backend: Python (NumPy, Open3D optional)
- Frontend: Web (React / Streamlit)
- Data: LiDAR dataset (.pcd / .bin)

5. Non-Functional Requirements

5.1 Performance

- Near real-time updates (target: 5–15 FPS)
- Efficient memory usage via adaptive grid

5.2 Usability

- Simple UI
- Quick parameter adjustment
- Visual clarity for judges

5.3 Reliability

- System should handle incomplete or noisy data
- Must not crash during demo

5.4 Scalability (Conceptual)

- Can be extended to real LiDAR systems
- Can integrate advanced ML models

6. Constraints

- Limited time (3 days)
- No real LiDAR hardware
- Use of simplified classification logic

7. Assumptions

- Dataset represents realistic environment
- Ultrasonic demo is symbolic, not precise
- Judges evaluate concept and execution

8. Future Enhancements

- Integration with real LiDAR sensors
- Deep learning-based segmentation
- Path planning module

- Autonomous navigation integration

9. Summary

FoveaDrive demonstrates adaptive perception inspired by human vision, efficient mapping using variable resolution, and vehicle-aware terrain analysis. The system prioritizes clarity, real-time visualization, and practical relevance over complexity.